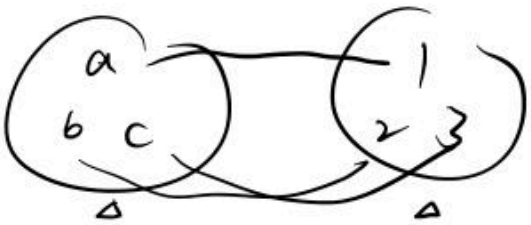
第五章 基数

2025年7月15日 14:40

5.1 集合的等势



① 等势

集合A与B等势 $\Leftrightarrow \exists f: A \to B$, f 为双射，记为 $A \approx B$

N, $N_{偶}$, $N_{奇}$, $N_{偶/奇} = \{n \mid n \in N \wedge n$ 为偶/奇数$\}$

$f: N \to N_{偶}$, $f(n)=2n$ / $f(n)=2n+1$

f 为双射

$N \approx N_{偶} \approx N_{奇}$

② $\begin{cases} Z \approx N \\ N \times N \approx N \\ N \approx Q \\ (0,1) \approx R \\ [0,1] \approx (0,1) \end{cases}$

1) $Z \approx N$

$$f(n)=\begin{cases} 0 & , n=0 \\ 2n & , n>0 \\ 2|n|-1 & , n<0 \end{cases}$$

$\mathbb{Z}$ ... -2 -1 0 1 2 ...

$\mathbb{Z} \to \mathbb{N}$ ... f(-2) f(-1) f(0) f(1) f(2) ...

$\mathbb{N}$ ... 3 1 0 2 4 ...

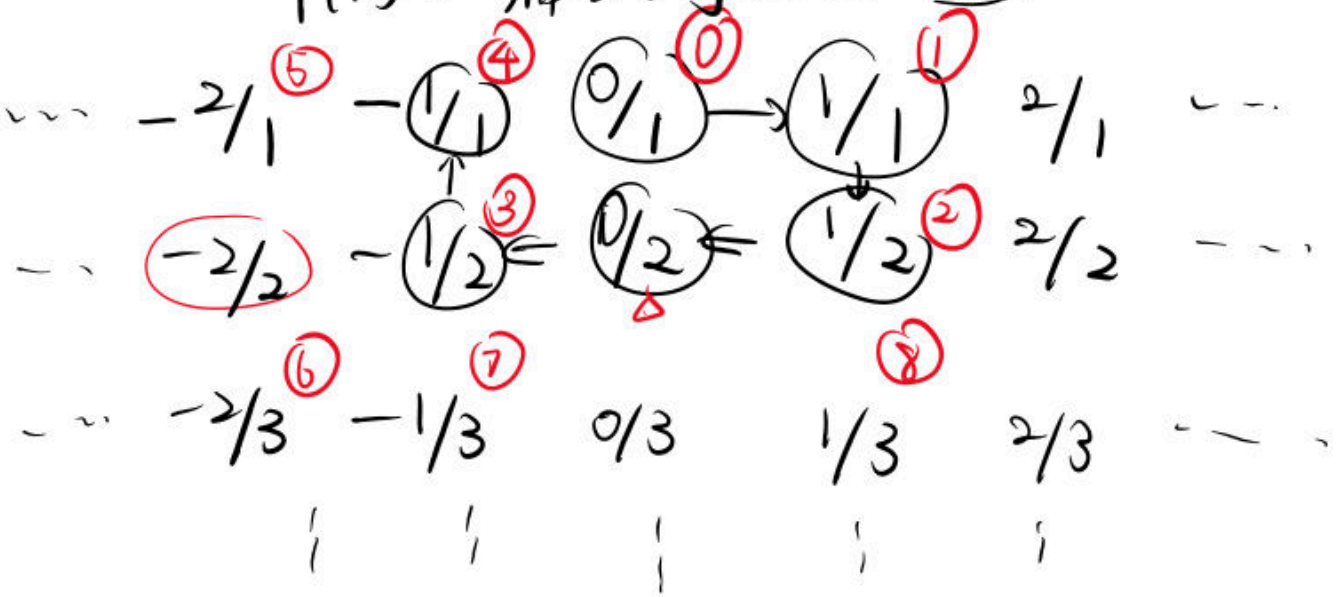
(2) $\mathbb{N} \times \mathbb{N} \simeq \mathbb{N}$

$f(\langle i,j \rangle) = 2^i(2j+1) - 1$

	$\langle 0,j \rangle$	$\langle 1,j \rangle$	$\langle 2,j \rangle$
$\langle i,0 \rangle$	0	1	3
$\langle i,1 \rangle$	2	5	11
$\langle i,2 \rangle$	4	9	19
$\langle i,3 \rangle$	6	13	27

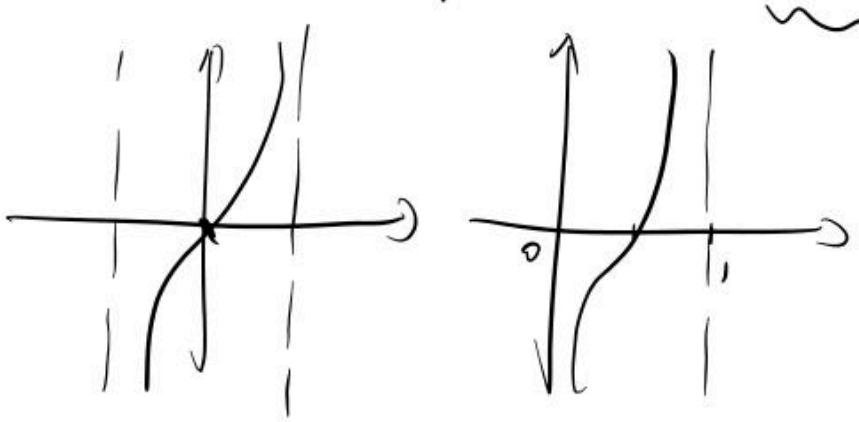
(3) $\mathbb{N} \simeq \mathbb{Q}$

$f(n) =$ 编号[n]的既约分数



(4) $(0,1) \approx \mathbb{R}$

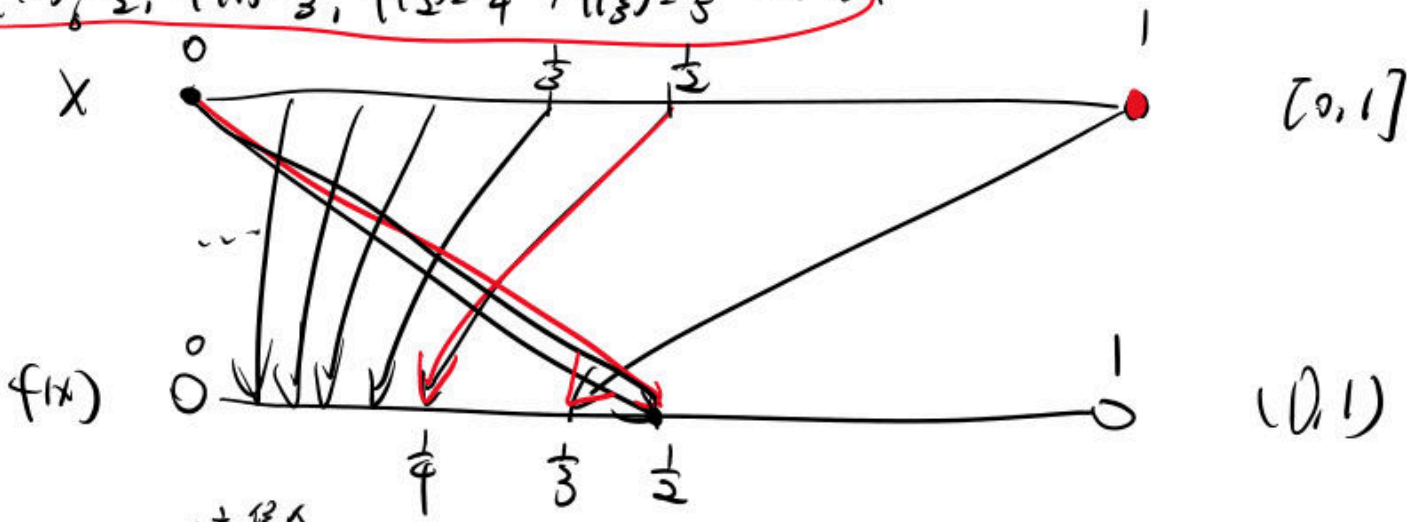
$f(x) = \tan[(x-\frac{1}{2})\pi]$



(5) $[0,1] \approx (0,1)$

$$f(x) = \begin{cases} 1/2, & x = 0 \quad ① \\ 1/(n+2), & x = 1/n, \ n \in \mathbb{N} - \{0\} \quad ② \\ x, & \text{else} \end{cases}$$

$f(0)=\frac{1}{2}$, $f(1)=\frac{1}{3}$, $f(\frac{1}{2})=\frac{1}{4}$, $f(\frac{1}{3})=\frac{1}{5}$ ...



对称性

③ $\begin{cases} A \approx A \\ A \approx B \Rightarrow B \approx A \\ A \approx B \wedge B \approx C \Rightarrow A \approx C \end{cases}$

(4) $[\ ] \simeq [\ ] \simeq (\ ) \simeq [\ ]$

④ $\begin{cases} N \approx Z \approx Q \approx N \times N \\ (0,1) \approx [0,1] \approx R \end{cases}$

⑤ 康托定理 $\begin{cases} N \not\approx R \\ A \not\approx P(A) \end{cases}$ ☆

⑥ 定理 $P(A) \approx 2^A = A \to \{0,1\}$

$\{f \mid f: A \to 2\}$

全体函数所组成的集合

⑦ 有穷集 / 无穷集

1,2,3 有穷集 ⇔ 与某个自然数等势的集合

⇔ (不能)与(自身的真子集)建立双射 ☆

N,Q,R,Z 无穷集 ⇔ 不能与某个自然数等势的集合

⇔ 能与(自身的真子集)建立双射

⑧ N 是无穷集

任何有穷集都与唯一的自然数等势

有穷集的子集仍为有穷集

5.2 康托定理，有穷集证明

2025年7月16日 14:45

① $\begin{cases} N \not\approx R \\ A \not\approx P(A) \end{cases}$

1) $N \not\approx R$

反证：假设存在 $N \approx R \approx [0,1] \Rightarrow N \approx [0,1]$

存在 $f: N \to [0,1]$，双射

$\forall n \in N, f(n) = X_{n+1}$

$\begin{cases} f(0) = X_1 \\ f(1) = X_2 \\ f(2) = X_3 \\ f(3) = X_4 \\ \vdots \quad \vdots \end{cases}$

值域：$\text{ran} f = [0,1] = \{X_1, X_2, X_3 \cdots\cdots\}$

$X_i = 0.a_1^{(i)} a_2^{(i)} a_3^{(i)} \cdots\cdots$

$\begin{cases} X_1 = 0.a_1^{(1)} a_2^{(1)} a_3^{(1)} \cdots \\ X_2 = 0.a_1^{(2)} a_2^{(2)} a_3^{(2)} \cdots \\ X_3 = 0.a_1^{(3)} a_2^{(3)} a_3^{(3)} \cdots \\ \vdots \quad \vdots \end{cases}$

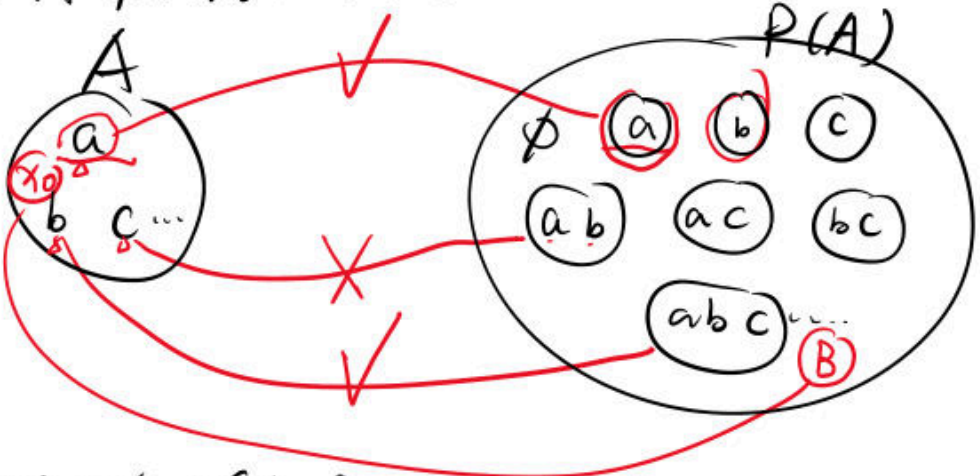
$X' = 0.b_1 b_2 b_3 \cdots$

(1) $0 \le b_j \le 9$，j 为 1, 2, 3, ……

(2) $b_n \neq a_n^{(n)}$

X' ∉ $\{x_1, x_2, x_3 \cdots\}$ 没有自然数与之对应

(2) $A \not\sim P(A)$，对任意集合 A



假设存在双射 $f: A \to P(A)$

i) 令 $B = \{x \in A \mid x \notin f(x)\} \in P(A)$

$x_0 \in A \mid x_0 \notin f(x_0)$

ii) 由于 f 为双射，存在 x_0，使得 $f(x_0) = B$

如果 $x_0 \in B \overset{i)}{\Longleftrightarrow} x_0 \notin f(x_0) \overset{f(x_0)=B}{\Longleftrightarrow} x_0 \notin B$ paradox

② 不存在与自己的真子集等势的自然数

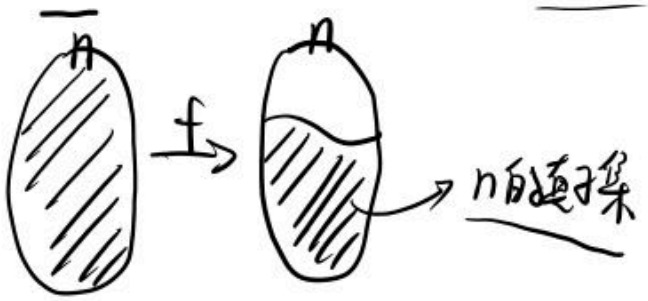
证明：令 $S = \{n \mid n \in N \wedge \forall f(f \in (n \to n) \wedge f \text{单射} \to f \text{满射})\}$

自己 → 自己

n → n

f 为单射 → 满射 ✓

→ 不满射 ✓



不满射 ✓

(1) $0 \in S$: $f \in (0 \to 0)$

f为空函数 $\Rightarrow$ f 满射

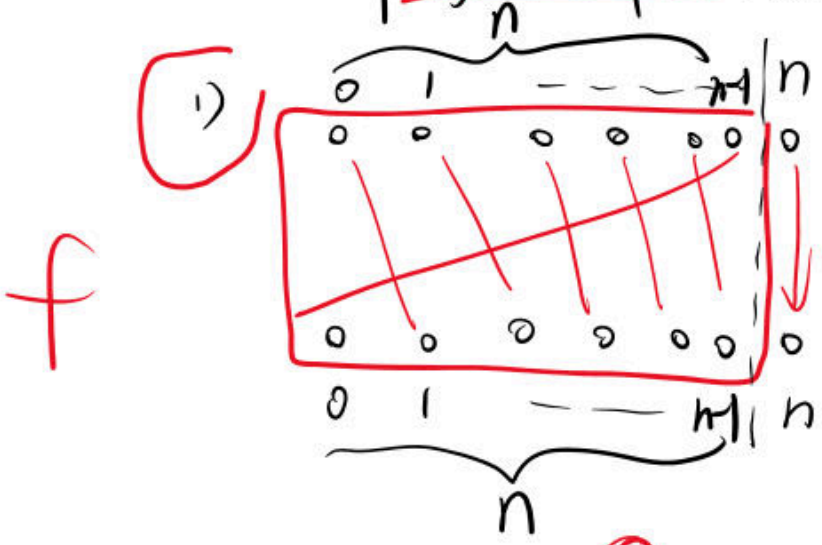
$n^+ = \{0, 1, 2 \cdots n\}$

$n = \{0, 1, 2 \cdots n-1\}$ ☆

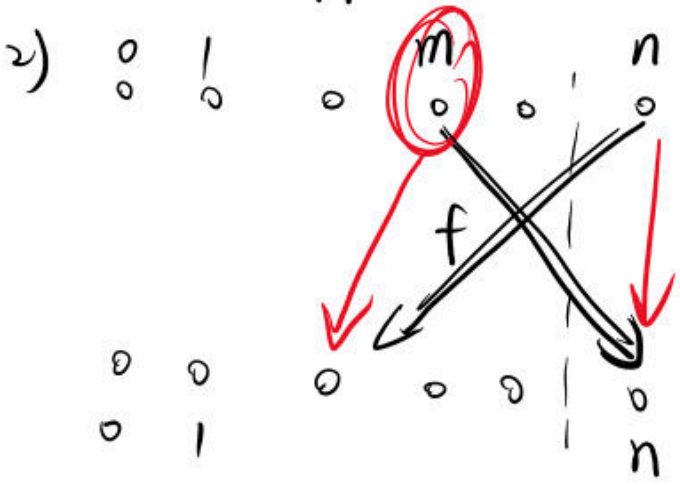
(2) $n \in S \Rightarrow n^+ \in S$

证明: $f: n^+ \to n^+$ 单射 $\Rightarrow$ f 满射

因为 f 任取的, { i) n在f下封闭 ii) n在f下不封闭



必有 $f(n) = n$



必有 $f(m) = n$, $m \in n$

$$h(x)=\begin{cases} f(n), & x=m \\ n, & x=n \\ f(x), & x\neq m \wedge x\neq n \end{cases}$$

$\therefore S=N$

§5.3 基数和序数的比较与运算

2025年7月22日 5:27

① 基数的定义

1) $A \approx B \Leftrightarrow \underline{\text{Card}\, A = \text{card}\, B}$

2) $A \approx n$ (有穷集 A) $\Leftrightarrow \text{Card}\, A = n$

3) $\text{card}\, N = \underline{\aleph_0}$

4) $\text{card}\, R = \underline{\aleph_1} = \aleph$

5) $\underbrace{0, 1, 2, \cdots\cdots}_{\text{有穷基数}}, \underbrace{\aleph_0, \aleph}_{\text{无穷基数}}$ 为基数

若 $\text{card}\, A = \aleph_i$, 则 $\text{card}\ P(A) = \aleph_{i+1}$

用 κ, λ, μ 表示基数 (任意)

② K_κ

设 κ 为任意基数, 令

$$\underline{K_\kappa} = \{x \mid x \text{为集合且} \text{Card}\, X = \kappa\}$$

当 $\kappa = 0$, $K_0 = \{\emptyset\}$ 为集合

当 $\kappa \neq 0$, $\underline{K_\kappa \text{不是集合, 是类}}$

③ eg (1) $A = \{a, b, c\}$

$B = \{\{a\}, \{b\}, \{c\}\}$

$card\ A = card\ B = 3$

(2) $N_{偶}$, $N_{奇}$

$card\ N_{偶} = card\ N_{奇} = \aleph_0$。

(3) $[0,1]$, $(0,1)$

$[0,1] \approx (0,1) \approx R$

$card\ [0,1] = card\ (0,1) = \aleph$

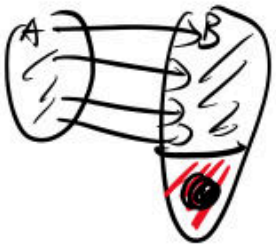
④ 优势劣势

B比A优势

$\Leftrightarrow$ A比B劣势

$\Leftrightarrow A \preceq\cdot B$

$\Leftrightarrow \exists f: A \to B$ 单射



绝对优势，绝对劣势

B比A绝对劣势

$\Leftrightarrow$ A比B绝对优势

$\Leftrightarrow A \prec\cdot B$

$\Leftrightarrow A \preceq\cdot B \wedge A \not\approx B$

⑤ 1) $A \preceq\cdot B \Leftrightarrow \exists C \subseteq B$，使得 $A \approx C$

2) 推论：$A \subseteq B \Rightarrow A \preceq\cdot B$

$$A \approx B \Rightarrow A \lesssim\cdot B \wedge B \lesssim\cdot A$$

3) $A \lesssim\cdot A$

$A \lesssim\cdot B \wedge B \lesssim\cdot C \Rightarrow A \lesssim\cdot C$

4) $A \lesssim\cdot B \wedge C \lesssim\cdot D$

$\Rightarrow$ (1) $A \cup C \lesssim\cdot B \cup D$ $(B \cap D = \emptyset)$

(2) $A \times C \lesssim\cdot B \times D$

5) $\operatorname{card} A = \operatorname{card} B = \kappa \wedge \operatorname{card} C = \operatorname{card} D = \lambda$

$\Rightarrow A \lesssim\cdot C \longleftrightarrow B \lesssim\cdot D$

⑥ 基数的比较

$\operatorname{Card} A = \kappa$, $\operatorname{card} B = \lambda$

$\kappa \leq \lambda \Leftrightarrow A \lesssim\cdot B$

$\kappa < \lambda \Leftrightarrow A <\cdot B$

$\kappa \leq \lambda \Rightarrow \exists A \subseteq B$, $\operatorname{card} A = \kappa \wedge \operatorname{card} B = \lambda$

$0 < 1 < 2 < 3 \cdots$

$0 \leq \kappa$ (链)

$n < \aleph_0 < \aleph_1 \cdots$

⑦

(1) $\operatorname{Card} A < \operatorname{card} P(A)$

(2) $K \leq K$

$K \leq \lambda \wedge \lambda \leq \mu \Rightarrow K \leq \mu$

(3) $\begin{cases} K \leq \lambda \wedge \lambda \leq K \Rightarrow K = \lambda \\ A \leq\cdot B \wedge B \leq\cdot A \Rightarrow A \approx B \text{ ★} \end{cases}$

Schroder-Bernstein 定理

(4) $A \subseteq B \subseteq C \wedge A \approx C \Rightarrow A \approx B \approx C$

(5) $R \approx (N \to 2) = 2^N = N \xrightarrow{f} \{0,1\}$

全体f函数的集合

⑧ 可数集

可数集(可列集): $\text{card} A \leq \aleph_0$

有穷可数集: $n, (\forall n \in N)$

无穷可数集: $N, Q, Z \cdots$ $(\text{card} A = \aleph_0)$

A为无穷集 $\Rightarrow$ P(A) 不是可数集

⑨ 基数运算的定义

设 K, λ 为基数, K, L 为集合, $\text{card} K = K$, $\text{card} L = \lambda$

(1) $K + \lambda = \text{card}(K \cup L)$, 其中 $K \cap L = \emptyset$

(2) $K \times \lambda = \mathrm{card}(K \times L) = K\lambda = K \cdot \lambda$

(3) $K^{\lambda} = \mathrm{card}(L \to K)$

(10) (1) $2^{\mathrm{card}\, A} = \mathrm{card}\, P(A)$

(2) $K < 2^{K}$

推论: (1) $\mathrm{card}\, P(N) = 2^{\aleph_0}$

(2) $\mathrm{card}\, P(R) = 2^{\aleph}$

(3) $\aleph = 2^{\aleph_0}$

$\Downarrow$

$R \approx P(N)$ ☆

(11) 设 K, λ, μ 为基数

(1) $K + \lambda = \lambda + K$

(2) $(K+\lambda)+\mu = K+(\lambda+\mu)$

(3) $K \cdot (\lambda+\mu) = K \cdot \lambda + K \cdot \mu$

(4) $K^{\lambda+\mu} = K^{\lambda} \cdot K^{\mu}$

(5) $(K \cdot \lambda)^{\mu} = K^{\mu} \cdot \lambda^{\mu}$

(6) $(K^{\lambda})^{\mu} = K^{\lambda \cdot \mu}$

(12) 设 $K \le \lambda, \mu$ 为基数

(1) $K+\mu \le \lambda+\mu$

(2) $K \cdot \mu \le \lambda \cdot \mu$

(3) $K^{\mu} \leq \lambda^{\mu}$

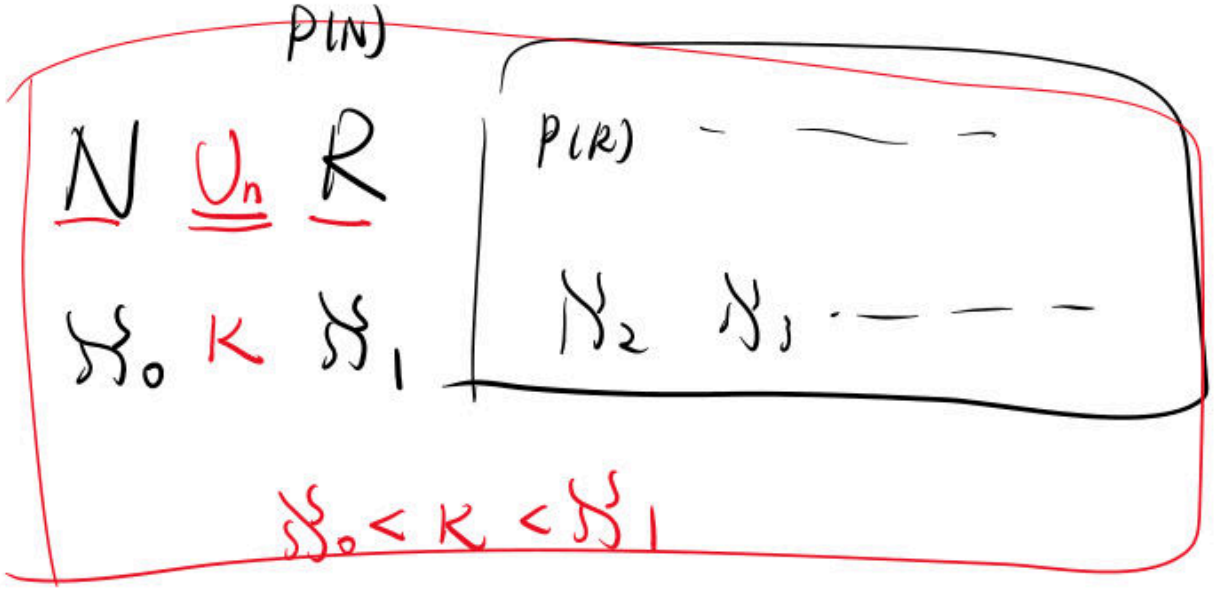
(4) $\mu^{K} \leq \mu^{\lambda}$，($K, \mu$ 不同时为0)

⑤ 无穷基数运算性质

设 K 为无穷基数，则 $K \cdot K = K$

λ 为基数，则 $K + \lambda = K \cdot \lambda = \max\{K, \lambda\}$ ($\lambda \neq 0$)

$$K^{K} = 2^{K}$$



5.4 Schroder - Bernstein 定理的证明

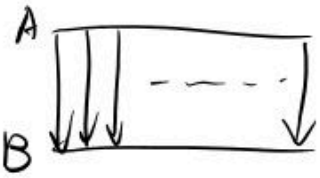
2025年7月22日 12:57

$A \leq\cdot B \wedge B \leq\cdot A \Rightarrow A \approx B$

证明:

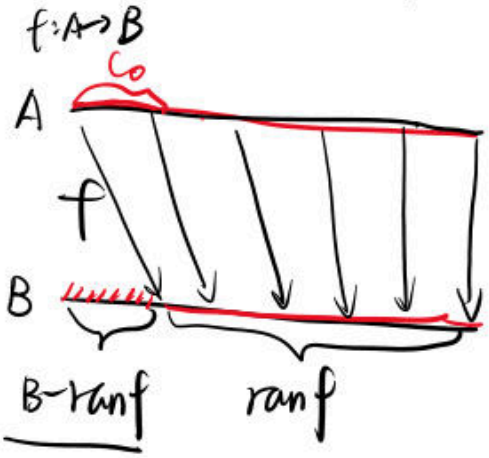
$A \leq\cdot B \wedge B \leq\cdot A$

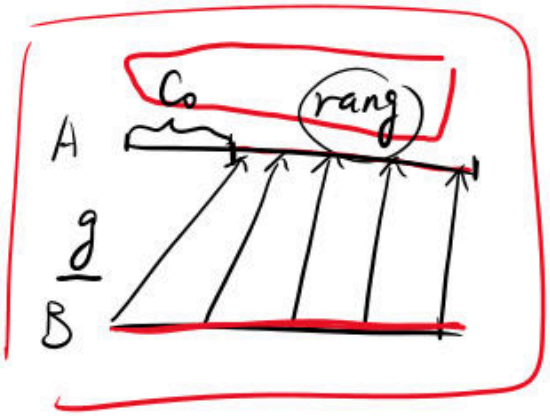
$\Rightarrow \exists f: A \to B$ 单射, $\exists g: B \to A$ 单射



1) 第一种情况 $\begin{cases} \operatorname{ran} f = B \\ \operatorname{ran} g = A \end{cases} \Rightarrow A \approx B$

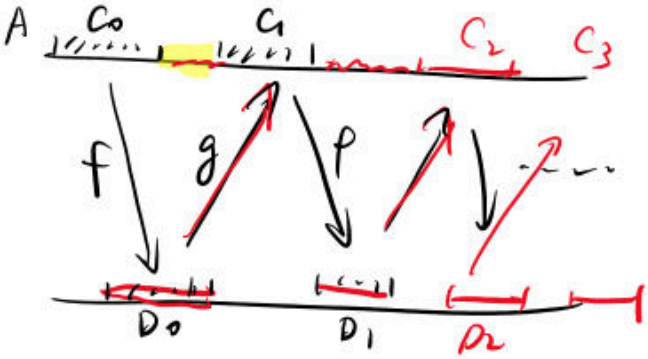
2) 第二种情况 $\begin{cases} A - \operatorname{ran} g \neq \emptyset & ① \\ B - \operatorname{ran} f \neq \emptyset \end{cases}$





令 $C_0 = A - \operatorname{ran} g \neq \emptyset$

$$\begin{cases} D_n = f(C_n) \\ C_{n+1} = g(D_n) \end{cases}, n \in N$$



$C = \cup \{C_n \mid n \in N\} \neq \emptyset \quad (C_0 \neq 0)$

$D = \cup \{D_n \mid n \in N\} = f(C) = g^{-1}(C - C_0)$

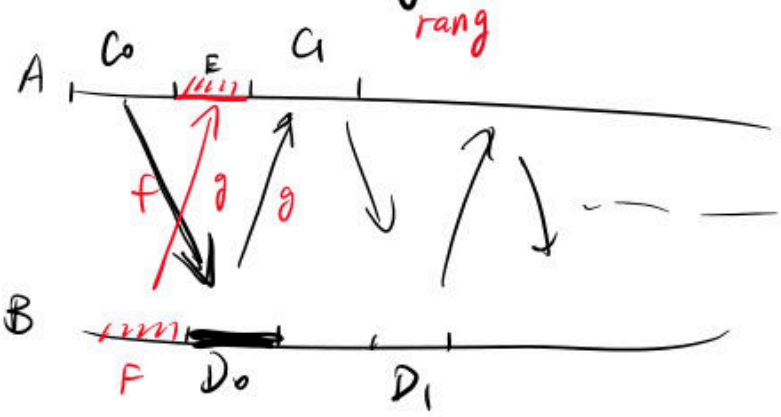
$E = A - C = (C_0 \cup \text{rang}) - (C_0 \cup (C - C_0))$

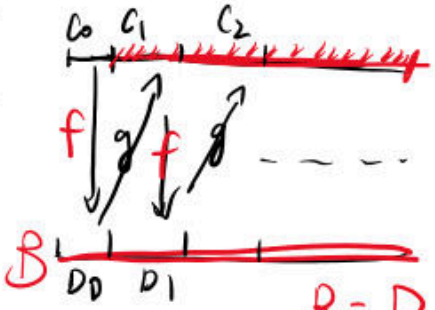
$= \text{rang} - (C - C_0) \neq \emptyset$

A

$C_0 \cup \text{rang} = C$

$\text{rang} = C - C_0$





$B = D$

$C_0 \neq \emptyset$

$f: A \longrightarrow B$ 单射

$g^{-1}: A - C_0 \longrightarrow B$ 单射

$A = A - C_0 \quad C = \emptyset$

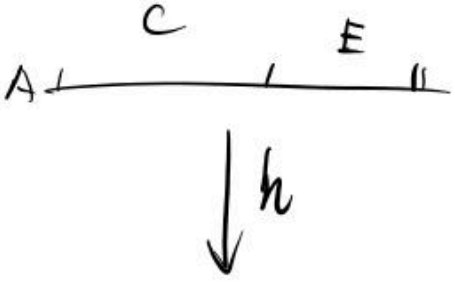
$C_0 \neq \emptyset$

$F = B - D = g^{-1}(g(B)) - g^{-1}(C - C_0)$

$= g^{-1}(g(B) - (C - C_0))$

$= g^{-1}(\text{rang} - (C - C_0)) = g^{-1}(E)$

$C \subset A$



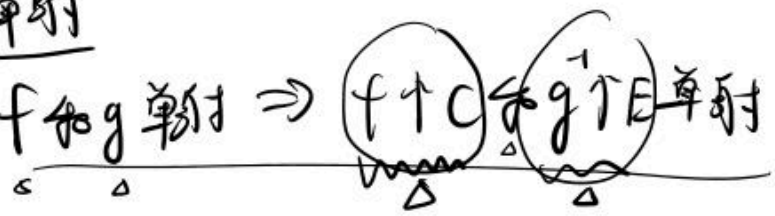
取 $h = f \upharpoonright C \cup g^{-1} \upharpoonright E$ (定义域)

$$h(x) = \begin{cases} f(x), & x \in C_n \\ g^{-1}(x), & \text{else} \end{cases}$$

$h(x) = \begin{cases} f(x), & \text{if } x \in C \\ g^{-1}(x), & \text{else} \end{cases}$



1) 单射



$\operatorname{ran}(f\restriction C) \cap \operatorname{ran}(g^{-1}\restriction E)$

$= f(C) \cap g^{-1}(E)$

$= g^{-1}(C - C_0) \cap g^{-1}(E) \subseteq g^{-1}(C) \cap g^{-1}(E)$

$= g^{-1}(C \cap E) = \emptyset$

$\therefore f\restriction C \cup g^{-1}\restriction E$ 为单射

(2) 满射

$B = D \cup F$

$= f(C) \cup g^{-1}(E)$

$= \operatorname{ran}(f\restriction C) \cup \operatorname{ran}(g^{-1}\restriction E)$

$= \operatorname{ran} h$

$\therefore h$ 为满射

5.5 序数与集合论公理

2025年7月22日 14:24

① 良序：$\langle A, < \rangle$ 自然数为良序，整数不是

任何非空子集都有最小元的序

$t_0 = \min(A)$

$t_1 = \min(A - \{t_0\})$

$t_2 = \min(A - \{t_0, t_1\})$

⋮

$t_0 < t_1 < t_2 \cdots$

② 序数

序数	良序集
0	$\emptyset$
1	$\{0\}$
2	$\{0,1\}$
3	$\{0,1,2\}$
⋮	⋮
w	全体自然数
$w+1$	
$w+2$	
$w+3$	
⋮	
$2w$	
$2w+1$	
$2w+2$	

$3\cdot\omega$

4ω

$\omega\cdot\omega=\omega^2$

ω^2+1

ω^2+2

$\omega^2+\omega$

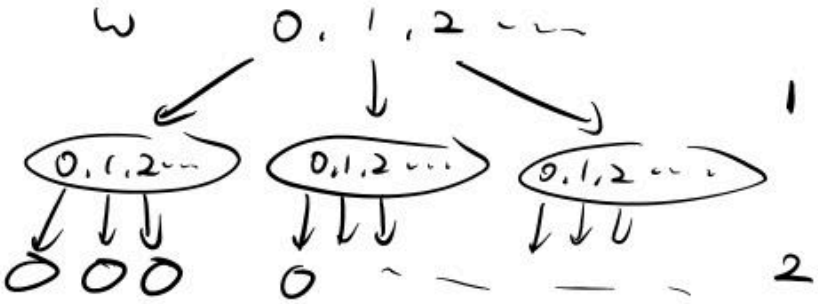
ω^3

ω^ω $(\omega+1)$次

ω^{ω^ω}

$\omega^{\omega^{\omega^{\cdots}}}$

ε_0



三类

$1, 2, \cdots \underline{\omega+1}, \omega+2 \cdots$ 有头有尾

$\omega, 2\omega, 3\omega, \omega^2, \omega^\omega$ 有头无尾

△ ○ ○ ○

③ 基数/序数

基数: 0, 1, 2 ⋯ | $\aleph_0$ | $\underline{2^{\aleph_0}}$ $\aleph^{2^{\aleph_0}}$

基数：0, 1, 2 ---- $\aleph_0$ ---- 2 ---- $2^{\aleph_0}$

序数：0, 1, 2 ---- ω, $\omega+1$ --- 2ω, --- ω^2 --- ω^ω ---- ω^{ω^ω}

初始序数：不与前面的序数等势的序数

0, 1, 2 ---, ω, ω^ω, ω^{ω^ω} ---

④ CH 连续统假设

$\neg \exists K (\aleph_0 < K < 2^{\aleph_0})$

$\aleph_1$ $R, P(N)$ → (A) ✓ ✗

$\aleph_0$ N, Q, Z ---

⑤ 公理系统 ZF, ZFC, ZFC+CH

1) ZF系统

① 外延公理：$A=B \Rightarrow \forall x(x\in A \leftrightarrow x\in B)$

② 无序对公理：a, b是集合 $\Rightarrow$ $\{a, b\}$是集合

③ 子集公理：A为集合 $\Rightarrow$ $\{x\in A \mid P(x)\}$为集合

④ 并集公理：A为集族 $\Rightarrow \bigcup A$是集合

⑤ 幂集公理：A为集合 $\Rightarrow P(A)$是集合

⑥ 空集公理：$\emptyset$为集合

⑦ 正则公理：A为非空集合 $\Rightarrow$ A有一个基础元素

(不属于A中其他元素的元素)

假设$(A\in A)$，集合A

① 构造 $S=\{A\}$

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② S有一个基础元素 x，存在 $x\in S$，$x\cap S=\emptyset$

③ $A\in S$，$A\in A$，$A=x$，$A\cap S=\emptyset$

$\Downarrow$

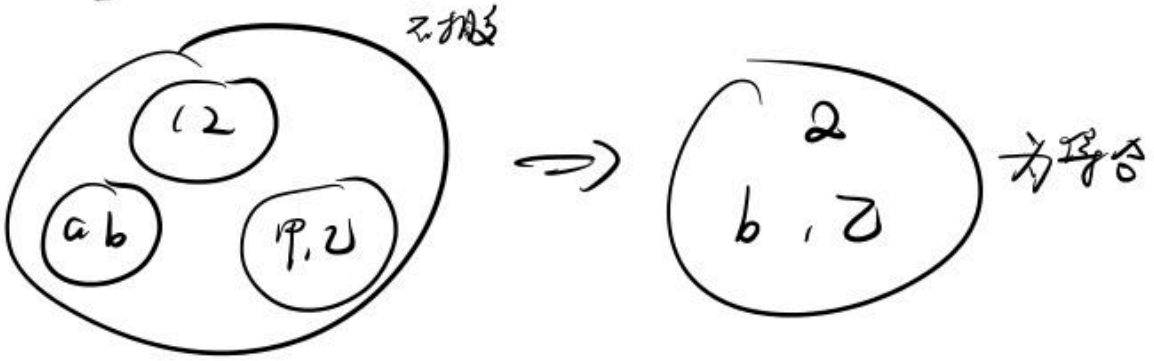
$A\in A\cap S$ ← 矛盾

⑧ 替换公理：f 为 A 上迭代 $\Rightarrow \{f(a)\mid a\in A\}$ 为集合

⑨ 无穷公理：N 为集合

⑥ ZFC 系统　　ZF + C

ZF模型



⑦ ZFC + CH 系统